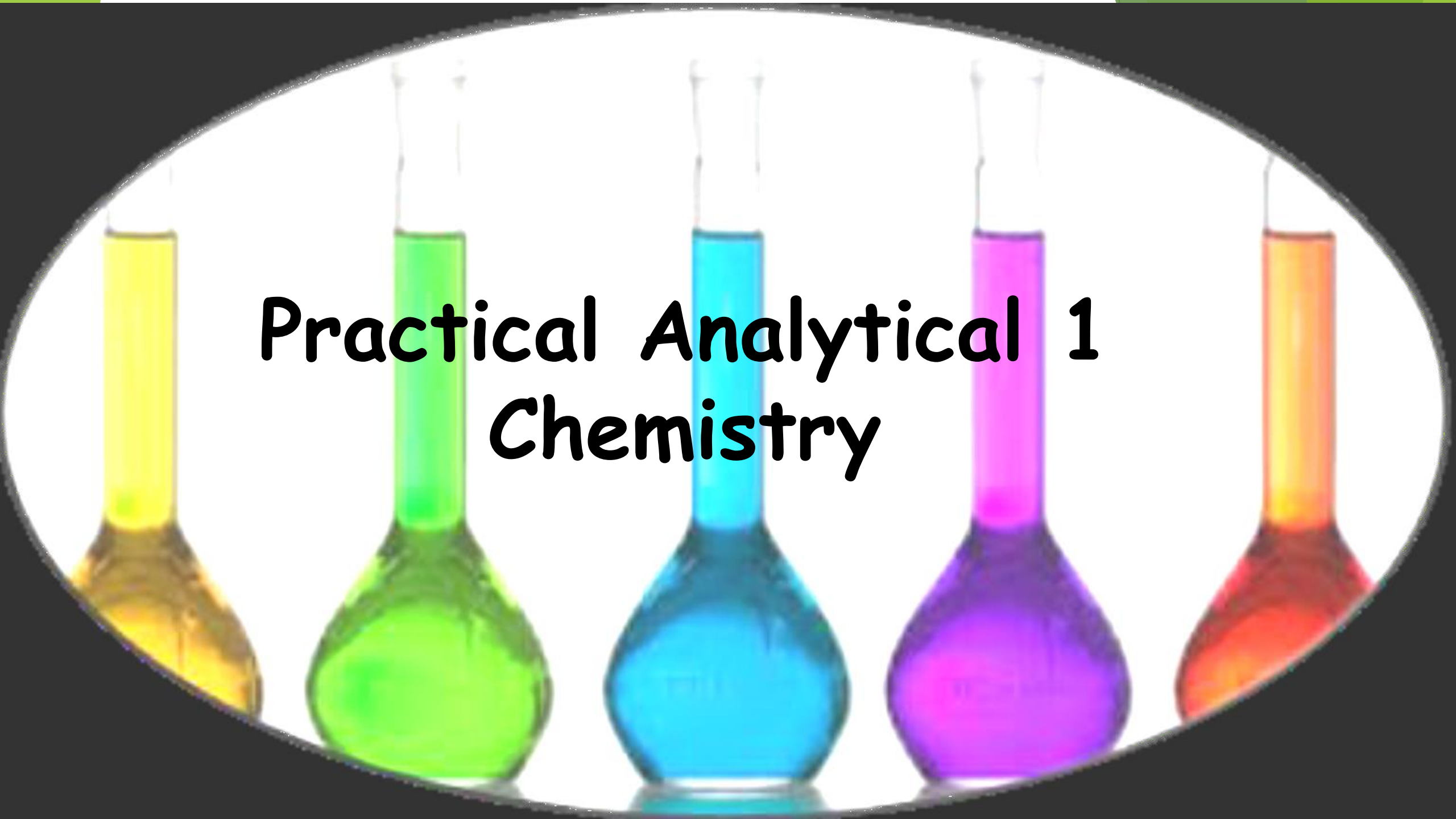
Practical Analytical 1 Chemistry



pH

- It is a quantitative measure of the acidity or basicity of aqueous or other liquid solutions.
- pH is defined by negative logarithm of the hydrogen ion concentration expressed by mol/L (Molarity).
- $pH = -\log[H^+]$.
- The pH is measured with (pH meter). Its reading is related to the potential difference between suitable electrodes (glace electrode and Ag/AgCl electrode) placed in the solution.

Rules

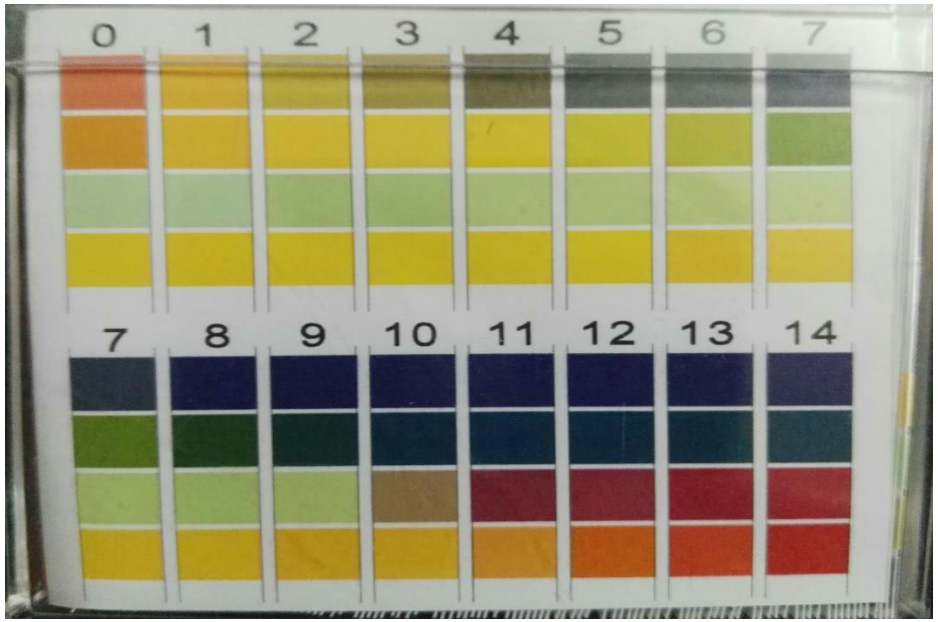
- pH + pOH= 14
- pH= -log[H^+]
- pOH= -log[^-OH]

for any solution

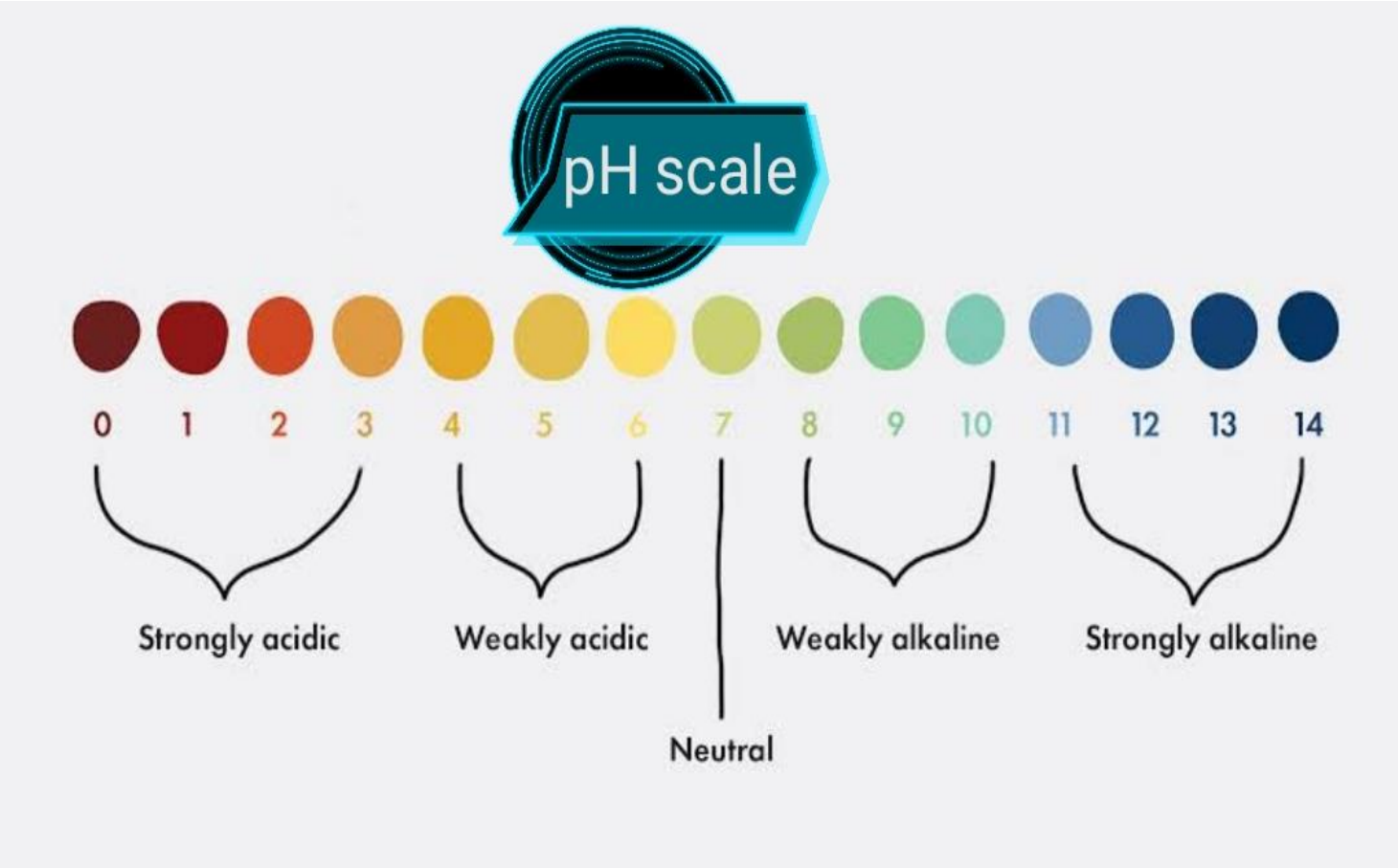
- k_a = [H^+] [A^-] /[HA] → for weak acid
- k_b = [B^+] [^-OH] /[BOH] → for weak base

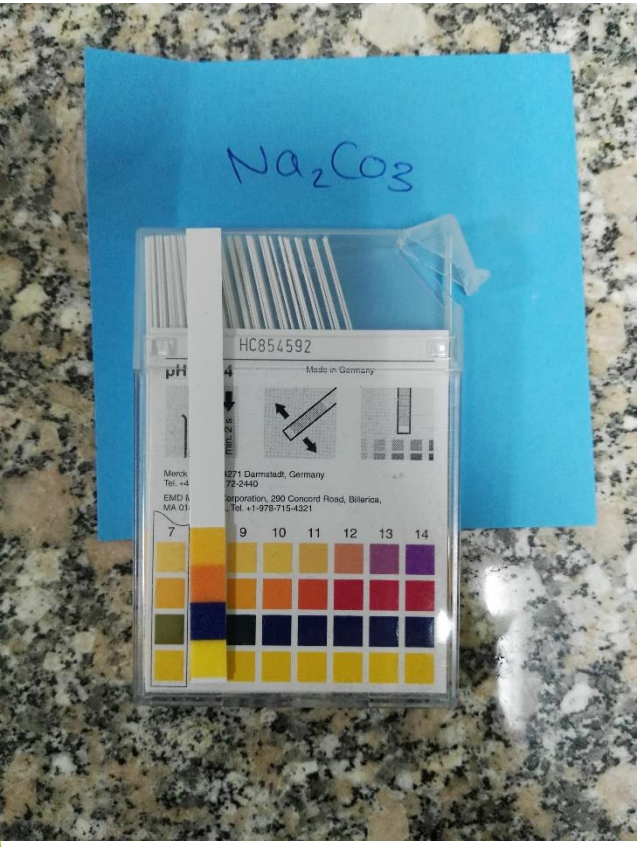
pH scale

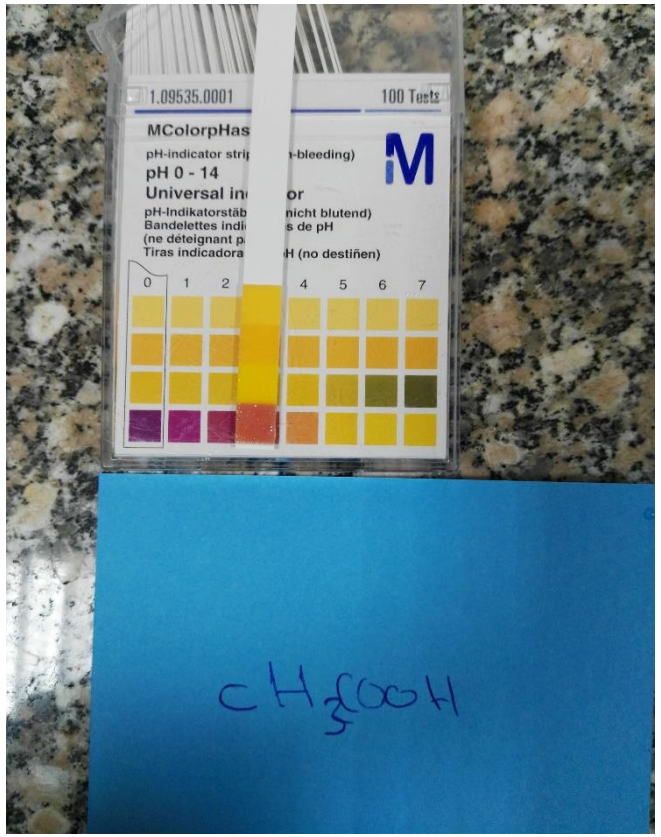
- The acidic pH values are < 7, and the basic (7< pH < 14) and the neutral pH is 7 .

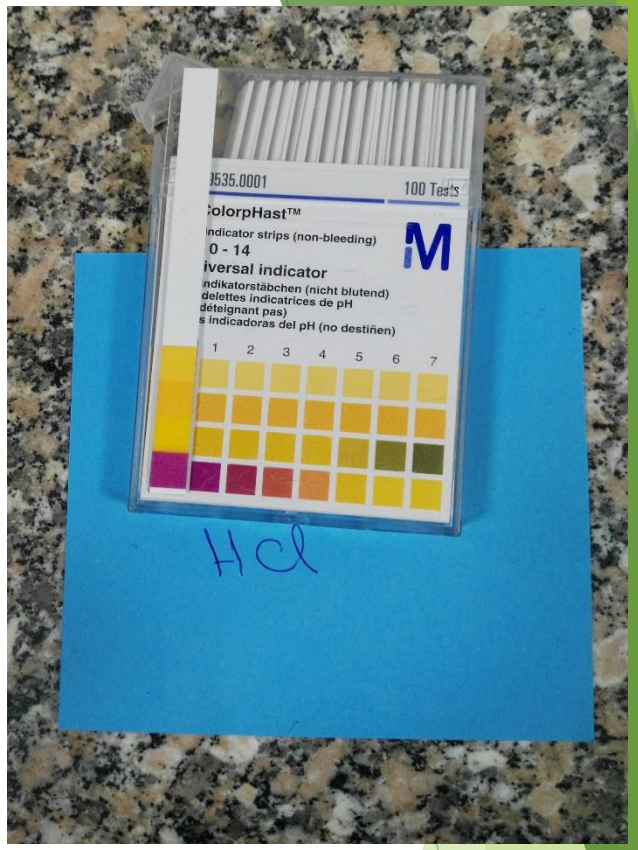


pH scale


pH scale
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14
Strongly acidic
Weakly acidic
Neutral
Weakly alkaline
Strongly alkaline



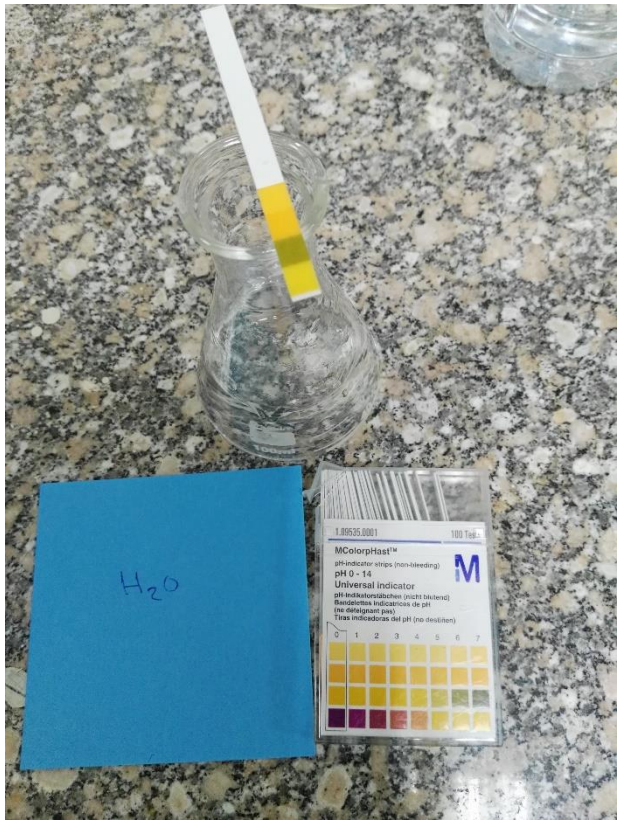




The pH value for Na_2CO_3 = 11, **(at 0.1 N)**

The pH value for CH_3COOH =3, **(at 0.1 N)**

The pH value for HCl =1, **(at 0.1 N)**



- The pH value for H_2O =7.

- The pH value for 0.1 N NaOH =13.

Example

- Calculate the pH of a solution of HCl with concentration of (10^{-10})M ? (i.e. $[H^+] = 10^{-10}$ M)
- the solution:

$pH = -\log[H^+]$

$pH = -\log[10^{-10}] = 10$

pOH

- It is defined by the negative logarithm of the hydroxyl ion concentration expressed in mol/L
- pOH= -log[$^{-}$OH].

- Note :

  pH + pOH= 14.

Example

- Calculate the pOH of a NaOH solution with a concentration of $(5 \times 10^{-5})M$? (i.e. $[OH^-] = 5 \times 10^{-5}$ M
- The solution

$pOH = -\log[OH^-]$

$pOH = -\log[5 \times 10^{-5}] = 4.3$

Examples

1) Calculate the pOH and pH of a solution of NaOH with a concentration of $(3 \times 10^{-2})M$?

<u>step1 :</u>

$pOH = -\log[^{-}OH] = -\log[3 \times 10^{-2}] = 1.5$

<u>step2 :</u>

$pH + pOH = 14$

$pH = 14 - pOH = 14 - 1.5 = 12.5$ → (strong base)

2) Calculate the pH and pOH of a solution of $Mg(OH)_2$ with a concentration of $(4.9 \times 10^{-9})M$?

step1 :

$pOH = -\log[^{-}OH] = -\log[(2)(4.9 \times 10^{-9})] = 8.01$

step2 :

$pH = 14 - pOH = 14 - 8.01 = 5.99$ → (weak acid)

Example for weak acid

- $k_a = [H^+][A^-]/[HA]$

- If the acid is not very diluted and not extremely weak (in other words if ions from water dissociation can be neglected) so $[H^+] = [A^-]$.

- $k_a = [H^+]^2/[HA]$

- Ex: Calculate the pH of acetic acid with a concentration of (0.1)M ,[K_a= 1.8 x 10^{-5}]?

$k_a = [H^+]^2 / [HA]$

$1.8 \times 10^{-5} = [H^+]^2 / 0.1$

$[H^+] = 1.3416 \times 10^{-3}$

$pH = -\log [1.3416 \times 10^{-3}] = 2.88$

The same thing can be done for calculating the pOH of a weak base.